Engineering Chemistry

UNIT – 3 : Electrochemistry and Applications

Cells, Nernst, titrations, sensors, batteries and fuel cells — the chemistry that powers your phone and your future, explained simply with clear diagrams.

Topics in this unit

1. Electrochemical cells & cell potential
2. Nernst equation & numericals
3. Potentiometry & conductometric titrations
4. Electrochemical sensors
5. Batteries: Zn-air & Li-ion
6. Fuel cells & PEMFC

* Study tip

Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. What is Electrochemistry?

Every time you switch on a phone, start a car or flick a torch, you are using electrochemistry — the branch of chemistry that links chemical reactions and electricity. It studies how a chemical reaction can produce an electric current, and how electricity can drive a chemical change.

DEFINITION

Electrochemistry: the study of the relationship between chemical energy and electrical energy — how spontaneous redox reactions generate electricity, and how electricity can force non-spontaneous reactions to happen.

Redox — the heart of it all

Every electrochemical change is a redox (reduction–oxidation) reaction, in which electrons move from one species to another.

- Oxidation = loss of electrons — happens at the anode.
- Reduction = gain of electrons — happens at the cathode.
- Memory hook: OIL RIG → Oxidation Is Loss, Reduction Is Gain.

NOTE

Oxidation and reduction always occur together — the electrons lost by one species are exactly the electrons gained by another. You can never have one without the other.

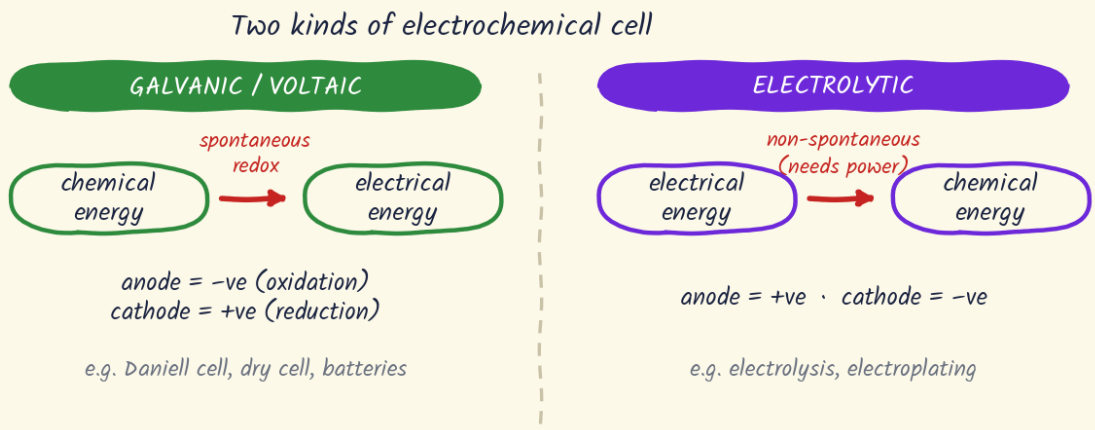


Fig 1 — A galvanic cell turns chemical energy into electricity (spontaneous); an electrolytic cell uses electricity to force a reaction.

2. Electrochemical Cells

An **electrochemical cell** is a device that either produces electricity from a chemical reaction, or uses electricity to carry one out. There are two families.

> DEFINITION
> **Galvanic (Voltaic) cell:** a cell that converts **chemical energy** into **electrical energy** using a **spontaneous** redox reaction. All batteries are galvanic cells.

How a galvanic cell works

- The two half-reactions are separated into two **half-cells**, so electrons are forced to travel through an external wire — that flow is the **current**.
- **Anode** (oxidation) is the **negative** terminal; **cathode** (reduction) is the **positive** terminal.
- A **salt bridge** (KCl or KNO_3) completes the circuit and keeps each solution electrically neutral.

The Daniell cell — a classic example

The Daniell cell uses a zinc electrode in $ZnSO_4$ and a copper electrode in $CuSO_4$. Zinc loses electrons more easily than copper, so zinc is the anode and copper is the cathode.

$$\text{Anode: } Zn \rightarrow Zn^{2+} + 2e^- \quad \text{Cathode: } Cu^{2+} + 2e^- \rightarrow Cu$$

Overall: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

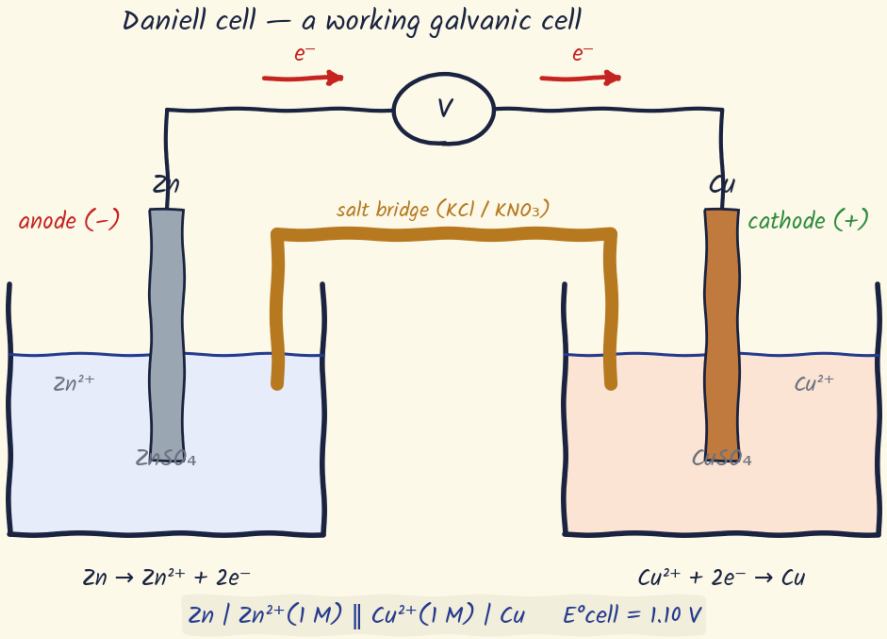


Fig 2 — The Daniell cell: electrons flow Zn → Cu through the wire while the salt bridge carries ions and keeps both beakers neutral.

> NOTE
> **Cell notation:** we write anode on the left, cathode on the right — $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$. A single bar | is a phase boundary (electrode–solution); the double bar ‖ is the salt bridge.

Electrolytic cell

DEFINITION

Electrolytic cell: a cell that uses **electrical energy** to drive a **non-spontaneous** chemical reaction (e.g. electrolysis, electroplating). Here the **anode is positive** and the **cathode is negative** — the opposite of a galvanic cell.

3. Electrode Potential & Cell Potential

Why do electrons flow from zinc to copper and not the other way? Because every electrode has a tendency to lose or gain electrons, measured as its **electrode potential**.

DEFINITION

Electrode potential: the tendency of an electrode to lose or gain electrons when dipped in its own ion solution. Measured against a reference, under standard conditions (1 M, 298 K, 1 atm) it is the **standard electrode potential (E°)**.

NOTE

Electrode potentials are measured against the **Standard Hydrogen Electrode (SHE)**, whose potential is fixed at $E^\circ = 0.00$ V by definition. Values are listed as standard **reduction** potentials — together they form the **electrochemical series**.

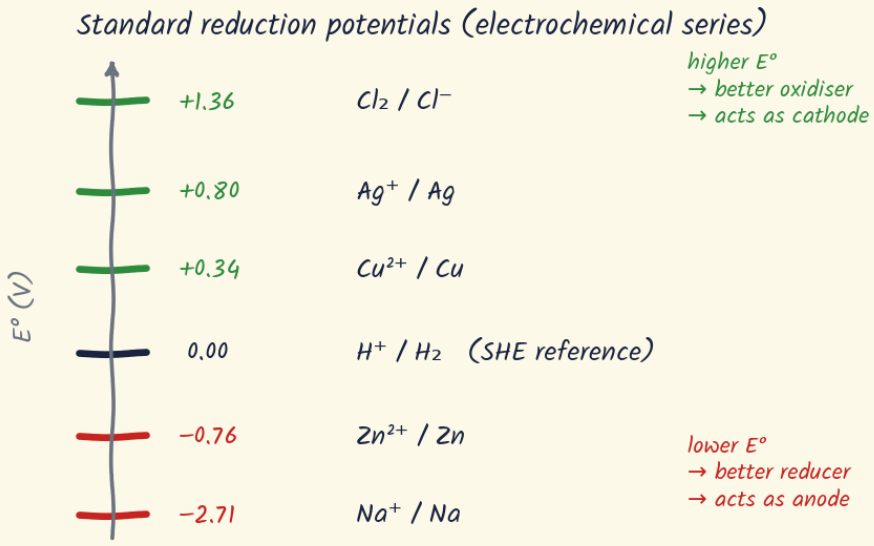


Fig 3 — The electrochemical series. The electrode with the higher E° acts as the cathode; the one with the lower E° acts as the anode.

Cell potential (EMF)

The cell potential or EMF is the voltage of the whole cell — the difference between the two electrode potentials. A positive E°cell means the reaction is spontaneous, so the cell can deliver current.

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

Both values are standard reduction potentials taken from the electrochemical series.

4. The Nernst Equation

Standard potentials assume every ion is at 1 M. In real cells the concentrations differ, and that changes the voltage. The Nernst equation tells us exactly how electrode and cell potentials depend on ion concentration and temperature.

For a single electrode

$$E = E^\circ + (0.0591/n) \log [M^{n+}] \text{ (at 298 K)}$$

n = number of electrons transferred; $[M^{n+}]$ = ion concentration

For the whole cell

$$E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q \text{ (at 298 K)}$$

Q = reaction quotient; R = 8.314 J K⁻¹mol⁻¹, F = 96500 C, T = 298 K are built into 0.0591

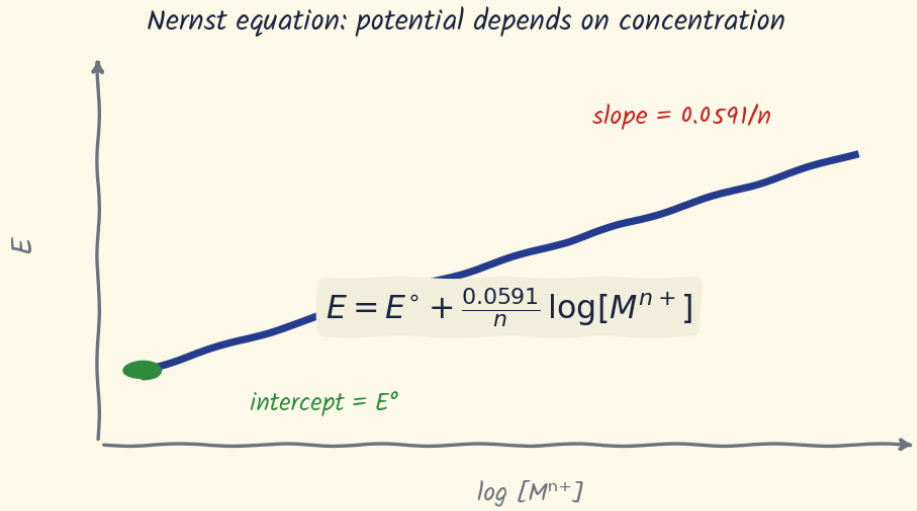


Fig 4 — Electrode potential varies linearly with log[ion]: the intercept is E° and the slope is 0.0591/n.

Worked examples

EXAMPLE 1

1. Standard EMF of the Daniell cell.

Given $E^\circ(Cu^{2+}/Cu) = +0.34$ V, $E^\circ(Zn^{2+}/Zn) = -0.76$ V.

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) = 1.10$ V.

EXAMPLE 2

2. EMF at non-standard concentration.

For $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, take $[Zn^{2+}] = 1.0$ M, $[Cu^{2+}] = 0.01$ M, $n = 2$, $T = 298$ K.

Step 1 — $Q = [Zn^{2+}] / [Cu^{2+}] = 1.0 / 0.01 = 100$.

Step 2 — $E_{cell} = 1.10 - (0.0591/2) \log(100) = 1.10 - (0.02955 \times 2)$.

Step 3 — $E_{cell} = 1.10 - 0.0591 = 1.04$ V. Lower $[Cu^{2+}]$ slightly lowers the EMF, exactly as Nernst predicts.

KEY POINT

When all ions are at 1 M, $\log Q = \log 1 = 0$, so $E_{cell} = E^{\circ}_{cell}$. The Nernst term only matters away from standard conditions.

5. Potentiometry & Potentiometric Titration

Instead of using a colour-change indicator, we can follow a titration by measuring voltage. This is potentiometry — and it is far more precise, especially for coloured or cloudy solutions.

DEFINITION

Potentiometric titration: a titration in which the EMF (or indicator-electrode potential) is measured after each addition of titrant. The equivalence point is found from a sharp jump in potential — no chemical indicator is needed.

- Plot E vs volume of titrant → an S-shaped (sigmoid) curve.
- Plot dE/dV vs volume → a sharp peak that pin-points the end point.
- Uses an indicator electrode (e.g. Pt) with a reference electrode (calomel).

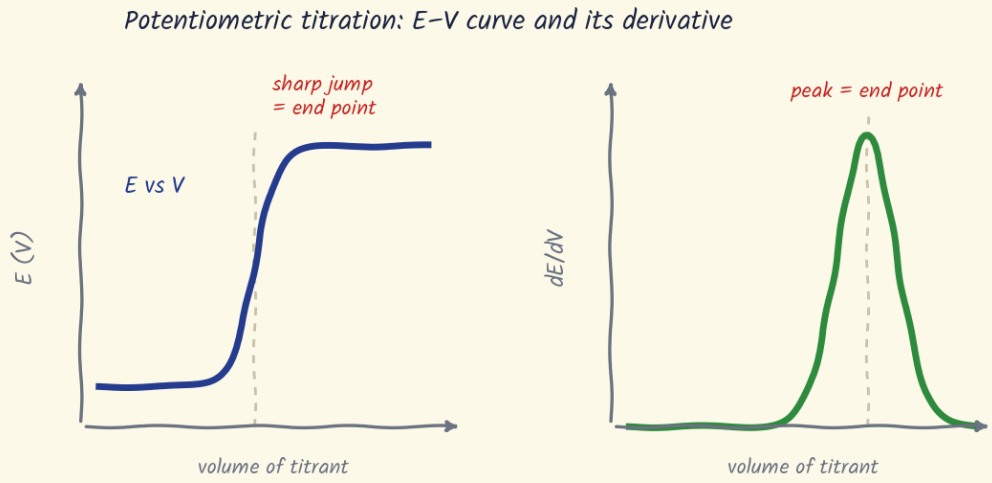


Fig 5 — The potential jumps sharply at the equivalence point; the derivative dE/dV peaks exactly there.

EXAMPLE

A redox titration of Fe^{2+} with $K_2Cr_2O_7$ (or Ce^{4+}) is followed with a Pt indicator electrode and a calomel reference. At the end point the electrode potential suddenly jumps as the last Fe^{2+} is used up.

NOTE

Advantages: no indicator needed, works for coloured/turbid solutions, very accurate, and can be automated.

6. Conductivity & Conductometric Titration

Ions carry current, so a solution's ability to conduct electricity depends on how many ions it has and how fast they move. Measuring this **conductance** is another neat way to find an end point.

DEFINITION

Specific conductance (κ): the conductance of a solution held between two electrodes 1 cm apart with 1 cm^2 area. Units: $S\ cm^{-1}$.

DEFINITION

Molar conductivity (Λ_m): the conducting power of all the ions from **1 mole** of electrolyte.

$$\Lambda_m = \kappa \times 1000 / C \text{ cell constant} = l / A$$

C = concentration (mol L^{-1}); l = distance between electrodes; A = electrode area

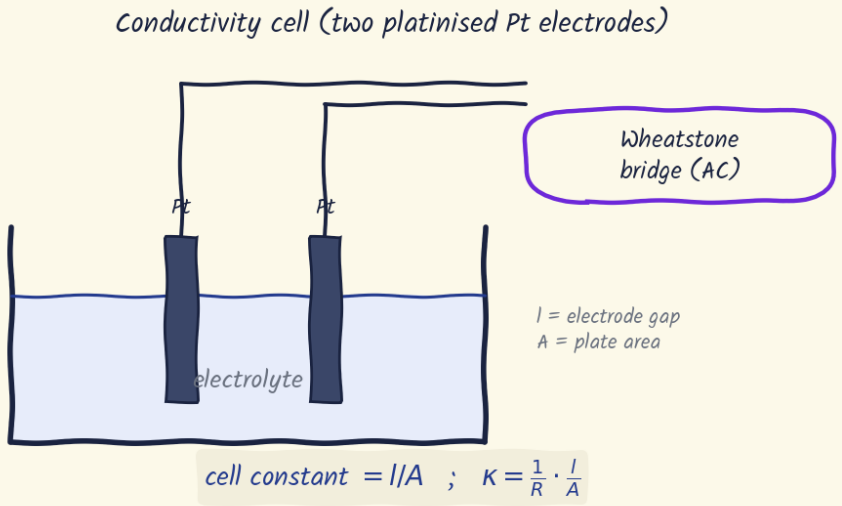


Fig 6 — A conductivity cell: two platinised Pt electrodes; resistance R is measured on an AC Wheatstone bridge.

Conductometric titration (strong acid vs strong base)

As we add NaOH to HCl, the conductance changes in a very telling way. The fast, highly-conducting H^+ ions are neutralised and replaced by slower Na^+ ions.

- **Before the end point:** conductance falls — fast H^+ replaced by slower Na^+.
- **At the end point:** conductance reaches a minimum (only salt + water left).
- **After the end point:** conductance rises again — excess fast OH^- ions are now added.

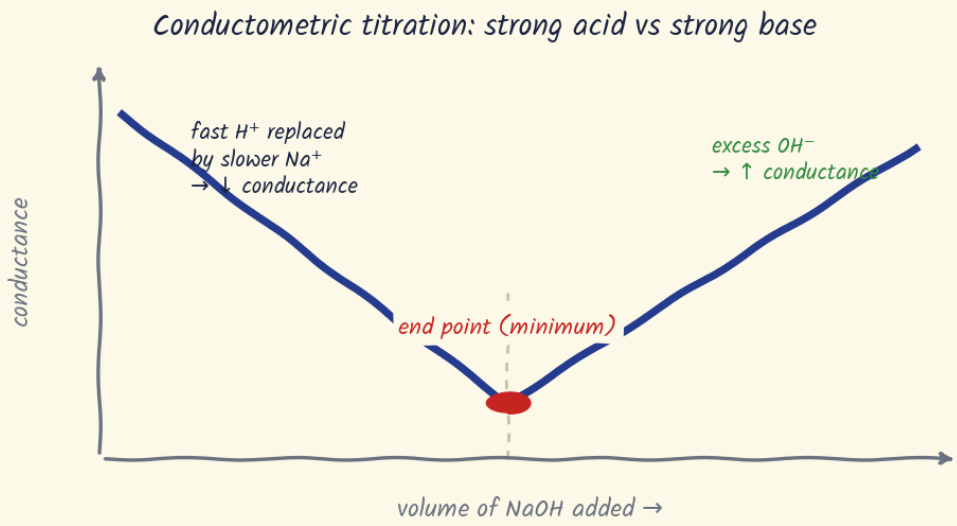


Fig 7 — The V-shaped curve: the end point is the minimum where the two straight lines meet.

KEY POINT

The end point is read from the break (minimum) of the V-shaped graph, not from any indicator — so even weakly coloured or dilute solutions can be titrated accurately.

7. Electrochemical Sensors

An electrochemical sensor turns the concentration of a chemical into an electrical signal we can read. They are cheap, fast and portable — think of the glucose meter used by diabetics.

DEFINITION

Electrochemical sensor: a device that converts a chemical quantity (concentration of an analyte) into a measurable electrical signal (voltage or current). The two main types are potentiometric and amperometric.

Potentiometric sensors

These measure a potential at (almost) zero current. The voltage is related to analyte concentration by the Nernst equation.

- pH glass electrode — potential depends on H^+ concentration.
- Ion-selective electrodes (ISE) — e.g. fluoride (LaF_3 crystal) and Ca^{2+} electrodes.

Amperometric sensors

These apply a fixed potential and measure the resulting current, which is directly proportional to analyte concentration.

- Clark oxygen electrode — measures dissolved O_2.
- Glucose biosensor — uses the enzyme glucose oxidase to measure blood glucose.

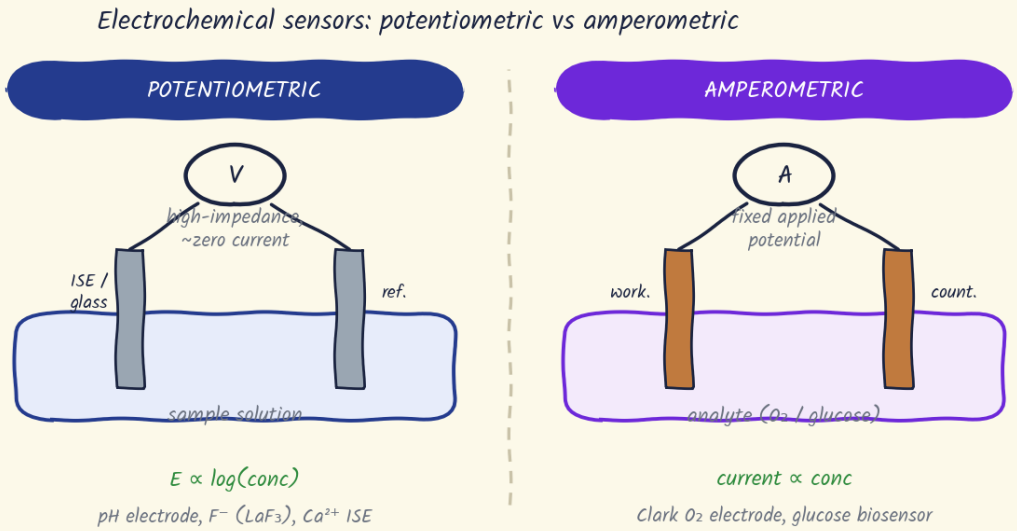


Fig 8 — Potentiometric sensors read a voltage at zero current ($E \propto \log c$); amperometric sensors read a current at a fixed voltage ($i \propto c$).

Feature	Potentiometric	Amperometric
Measured	potential (voltage)	current
Condition	~zero current	fixed applied potential
Relation	$E \propto \log(\text{conc})$ (Nernst)	current ∝ conc

Feature	Potentiometric	Amperometric
Examples	pH, F^- (LaF_3), Ca^{2+} ISE	Clark O_2, glucose sensor

8. Batteries: Zinc–Air & Li-ion

A battery is just one or more galvanic cells packaged to store and deliver electricity. Primary batteries are used once; secondary batteries can be recharged.

DEFINITION

Primary vs secondary: Primary cells give current until the reactants run out and are then thrown away (non-rechargeable). Secondary cells use a reversible reaction, so they can be recharged many times.

Zinc–air battery (primary)

A high-energy primary cell: a zinc anode, oxygen from the air as the cathode reactant on a porous carbon electrode, and a KOH electrolyte.

$$\text{Anode: } Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$$

Cathode: $\tfrac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$; Overall: $Zn + \tfrac{1}{2}O_2 \rightarrow ZnO$

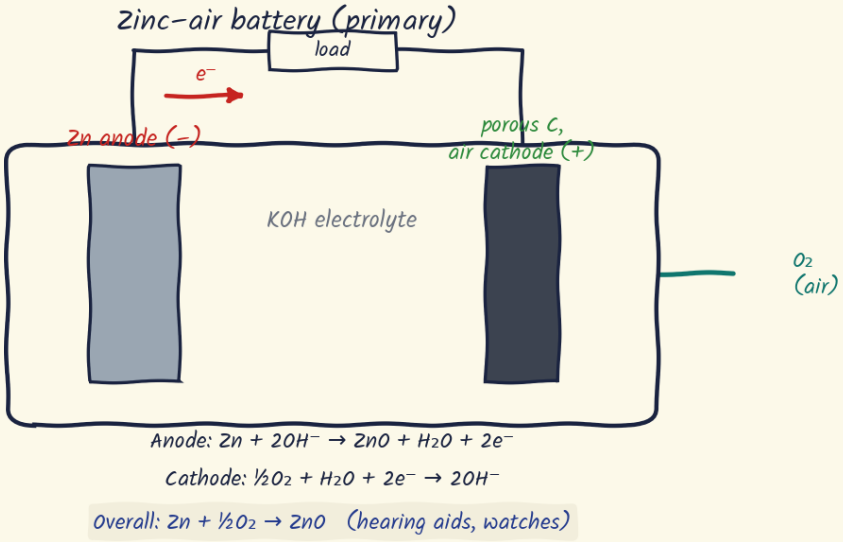


Fig 9 — Zinc–air cell: Zn is oxidised at the anode while oxygen from the air is reduced at the porous carbon cathode.

- **Very high energy density** because oxygen comes free from the air.
- **Uses:** hearing aids and watches.

Lithium-ion battery (secondary)

The rechargeable battery in your phone and laptop. The anode is **graphite** (LiC_6), the cathode is $LiCoO_2$, and the electrolyte is a lithium salt ($LiPF_6$) in an organic solvent with a separator.

> **NOTE**
> It is called a **"rocking-chair"** cell because Li^+ ions simply shuttle back and forth between the two electrodes on charge and discharge.

$$\text{Discharge anode: } LiC_6 \rightarrow Li^+ + e^- + 6C$$

Discharge cathode: $Li^+ + CoO_2 + e^- \rightarrow LiCoO_2$ (charging reverses both)

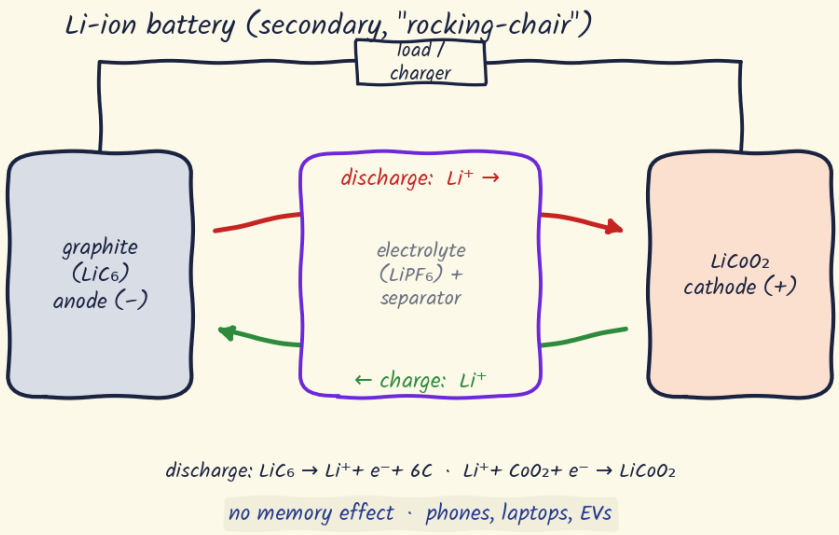


Fig 10 — Li-ion cell: Li^+ moves anode → cathode on discharge and back again on charging.

- **Uses:** phones, laptops and electric vehicles (EVs); **no memory effect**.

9. Fuel Cells

A **fuel cell** is like a battery that never runs down — as long as you keep feeding it fuel and oxygen, it keeps making electricity, cleanly and efficiently.

> DEFINITION
> **Fuel cell:** a device that converts the **chemical energy of a fuel** directly and continuously into **electricity** as long as fuel and an oxidant are supplied. It gives **high efficiency** and **low pollution**.

H_2–O_2 (alkaline) fuel cell

Hydrogen is the fuel and oxygen the oxidant, with a **KOH** electrolyte and **porous carbon** electrodes containing a **Pt / Ni** catalyst.

$$\text{Anode: } 2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$$

Cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; Overall: $2H_2 + O_2 \rightarrow 2H_2O$

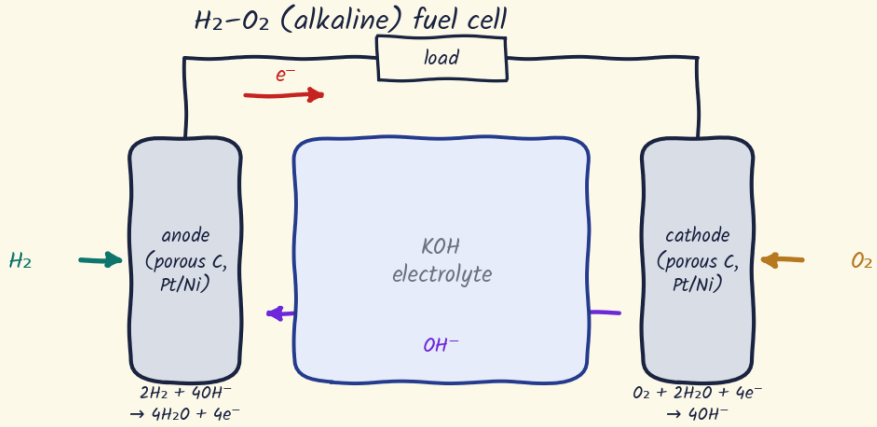


Fig 11 — H_2–O_2 alkaline fuel cell: the only product is water. Famously used in the Apollo spacecraft.

PEM fuel cell (PEMFC)

The Proton Exchange Membrane fuel cell replaces the liquid electrolyte with a solid polymer membrane (Nafion) that conducts H^+ ions. It runs cool (~60–80 °C) with a Pt catalyst.

$$\text{Anode: } H_2 \rightarrow 2H^+ + 2e^-$$

Cathode: $\tfrac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$ (H^+ crosses the membrane)

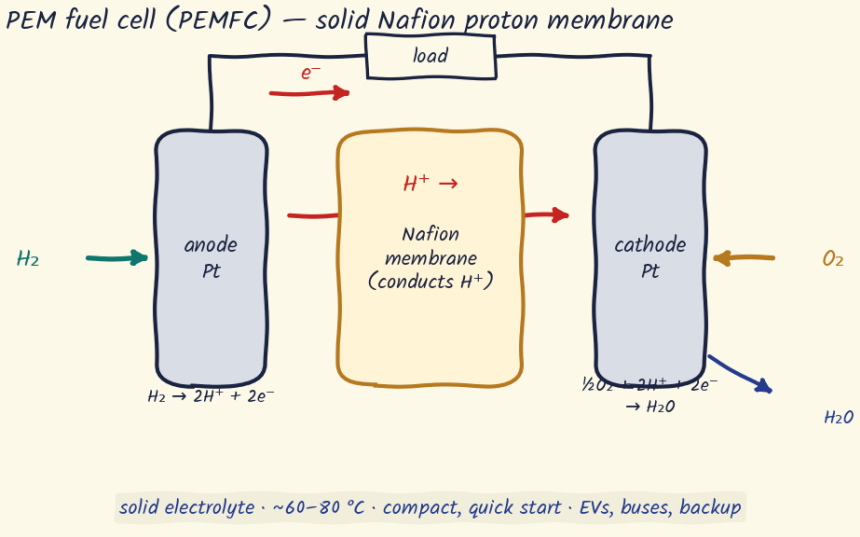


Fig 12 — PEMFC: protons cross the solid Nafion membrane while electrons drive the external load.

- **Advantages:** solid electrolyte (no leaks), compact and light, quick start-up.
- **Uses:** electric vehicles and buses, portable and backup power.

10. Applications & Importance

Electrochemistry quietly powers modern life:

- **Energy storage:** dry cells, Zn–air, lead-acid and Li-ion batteries for phones, laptops and EVs.
- **Clean power:** H_2–O_2 and PEM fuel cells for spacecraft, vehicles and backup supplies.
- **Measurement & sensing:** potentiometric/conductometric titrations, pH meters, ion-selective electrodes, Clark O_2 and glucose biosensors.
- **Industry:** electroplating, electro-refining of metals and corrosion protection.

QUICK RECAP

- Electrochemistry links chemical and electrical energy through redox reactions (oxidation at the anode, reduction at the cathode).
- A galvanic cell (e.g. Daniell, $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$) makes electricity spontaneously; an electrolytic cell uses electricity.
- $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$; a positive value means a spontaneous cell (Daniell = +1.10 V).
- Nernst: $E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q$ — potential depends on concentration.
- Potentiometric titration → sigmoid E–V with a sharp jump; conductometric (strong acid–base) → V-shaped curve, end point at the minimum.
- Sensors: potentiometric (pH, ISE; $E \propto \log c$) and amperometric (Clark O_2, glucose; $i \propto c$).
- Batteries: Zn–air (primary), Li-ion (secondary, rocking-chair). Fuel cells: H_2–O_2 alkaline and PEMFC (solid Nafion membrane).

EXAM-ORIENTED QUESTIONS

Q1. Differentiate between a galvanic cell and an electrolytic cell. [3 marks]

Ans. A galvanic cell converts chemical energy to electrical energy using a spontaneous redox reaction (anode −, cathode +). An electrolytic cell uses electrical energy to drive a non-spontaneous reaction (anode +, cathode −).

Q2. Describe the construction and working of the Daniell cell with its cell notation. [5 marks]

Ans. A Zn electrode in $ZnSO_4$ (anode) and a Cu electrode in $CuSO_4$ (cathode) joined by a salt bridge. $Zn \rightarrow Zn^{2+} + 2e^-$; $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons flow Zn → Cu externally. Notation: $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$.

Q3. State the function of a salt bridge. [2 marks]

Ans. It completes the internal circuit and maintains electrical neutrality in both half-cells by allowing ions (from KCl/KNO_3) to migrate.

Q4. Write the Nernst equation for a cell and explain each term. [3 marks]

Ans. $E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q$ at 298 K, where E°_{cell} = standard EMF, n = electrons transferred, Q = reaction quotient. It gives the EMF at non-standard concentrations.

Q5. Calculate the standard EMF of the Daniell cell. [2 marks]

Ans. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) = +1.10$ V.

Q6. How is the equivalence point found in a potentiometric titration? [5 marks]

Ans. The cell EMF is measured after each addition of titrant and plotted against volume, giving a sigmoid curve with a sharp jump at the equivalence point. A dE/dV vs V plot shows a peak exactly at the end point. No indicator is required.

Q7. Draw and explain the conductometric titration curve of a strong acid vs a strong base. [5 marks]

Ans. Conductance first falls as fast H^+ ions are replaced by slower Na^+, reaching a minimum at the end point; then it rises as excess fast OH^- is added — giving a V-shaped curve. The end point is the minimum.

Q8. Distinguish between potentiometric and amperometric sensors with examples. [3 marks]

Ans. Potentiometric sensors measure a potential at ~zero current, related to concentration by the Nernst equation (pH glass electrode, F^-/Ca^{2+} ISE). Amperometric sensors measure current at a fixed applied potential, proportional to concentration (Clark O_2 electrode, glucose biosensor).

Q9. Explain the working of a zinc–air battery with electrode reactions. [5 marks]

Ans. A primary cell: Zn anode, O_2 from air reduced on a porous carbon cathode, KOH electrolyte. Anode: $Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$; cathode: $\tfrac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$; overall $Zn + \tfrac{1}{2}O_2 \rightarrow ZnO$. Used in hearing aids and watches.

Q10. Why is a Li-ion battery called a "rocking-chair" cell? Give its electrodes. [3 marks]

Ans. Because Li^+ ions shuttle back and forth between the electrodes on charge and discharge. Anode = graphite (LiC_6), cathode = $LiCoO_2$, electrolyte = $LiPF_6$ in organic solvent. Rechargeable, no memory effect.

Q11. What is a fuel cell? Give the reactions of the H_2–O_2 (alkaline) fuel cell. [5 marks]

Ans. A cell that converts the chemical energy of a fuel directly and continuously to electricity while fuel + oxidant are supplied. Anode: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$; cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; overall $2H_2 + O_2 \rightarrow 2H_2O$. Used in Apollo spacecraft.

Q12. Write a short note on the PEM fuel cell (PEMFC). [3 marks]

Ans. It uses a solid proton-exchange membrane (Nafion) that conducts H^+, a Pt catalyst and runs at ~60–80 °C. Anode: $H_2 \rightarrow 2H^+ + 2e^-$; cathode: $\tfrac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Compact, solid electrolyte, quick start; used in EVs and backup power.